

Large Modular Congruences for which $x \mid 36n^3 - 19$

A CRT / cube-root construction with 30- and 40-digit moduli

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Abstract

We give a fully rigorous, arbitrarily-scalable construction of congruence classes $n \equiv a_1 \pmod{p_1}$ and (more generally) $n \equiv A \pmod{M}$ such that $36n^3 - 19$ is divisible by a prescribed modulus M built from primes of any prescribed size. The construction rests on the elementary fact that cubing is a bijection modulo any prime $p \equiv 2 \pmod{3}$. We instantiate this with 30-digit and 40-digit primes, verify every claim by direct computation (both symbolically for small primes and numerically for the large instances), and exhibit an explicit family in which $36n^3 - 19$ is divisible by a 120-digit modulus, itself a product of three 40-digit primes. All numerical claims below were checked by exact (arbitrary precision) integer arithmetic.

1 Problem restatement and corrected scope

We ultimately seek integers n, x, y , with $x \neq 0$, satisfying

$$y^2 = (6n + x)^2 + \frac{36n^3 - 19}{x}.$$

A necessary condition is $x \mid 36n^3 - 19$. The CRT construction in this note solves that divisibility condition in the following precise, uniform sense: it constructs a modulus M and residue $A \pmod{M}$ such that

$$M \mid 36n^3 - 19 \quad \text{for every } n \equiv A \pmod{M}.$$

We then make the structurally valid specialization

$$x = M, \quad n = A + Mt.$$

This guarantees the fraction is integral for every $t \in \mathbb{Z}$.

It is important not to infer more than this. Two independent congruences

$$n \equiv a_1 \pmod{p_1}, \quad x \equiv a_2 \pmod{p_2}$$

do not generally imply $x \mid 36n^3 - 19$: the first can control divisibility by p_1 , while the second controls a polynomial evaluated at x , not divisibility of the polynomial evaluated at n . Thus the variable class $x = p_2 k_2 + a_2$ is not used below as a consequence of CRT.

The construction is arbitrarily scalable: choose distinct primes $p_i \equiv 2 \pmod{3}$, solve $36a_i^3 \equiv 19 \pmod{p_i}$, combine the roots by CRT, and take $M = \prod_i p_i$. Larger moduli are easy to manufacture, but size alone is not the selection objective; Section 3 uses the arithmetic of the resulting elliptic curves to prioritize moduli.

2 The key lemma

Lemma 1. *Let p be a prime with $p \equiv 2 \pmod{3}$. Then the map $t \mapsto t^3 \pmod{p}$ is a bijection on $\mathbb{Z}/p\mathbb{Z}$. Consequently, for any $c \in (\mathbb{Z}/p\mathbb{Z})^\times$ there is a unique $a \in \{0, 1, \dots, p-1\}$ with $a^3 \equiv c \pmod{p}$, given explicitly by*

$$a \equiv c^e \pmod{p}, \quad e \equiv 3^{-1} \pmod{p-1}.$$

Proof. Since $p \equiv 2 \pmod{3}$ we have $p-1 \equiv 1 \pmod{3}$, so $\gcd(3, p-1) = 1$. Hence 3 is invertible modulo $p-1$; let $e = 3^{-1} \pmod{p-1}$. For $t \in (\mathbb{Z}/p\mathbb{Z})^\times$, Fermat's little theorem gives $(t^3)^e = t^{3e} = t^{1+k(p-1)} = t$ for the appropriate integer k , so cubing and raising to the power e are mutually inverse bijections on the multiplicative group; extending to $t = 0$ is trivial. $\square$

Construction 1. *Fix a prime $p \equiv 2 \pmod{3}$. Apply Lemma 1 with $c \equiv 19 \cdot 36^{-1} \pmod{p}$ to obtain the unique*

$$a \equiv (19 \cdot 36^{-1})^e \pmod{p}, \quad e = 3^{-1} \pmod{p-1},$$

satisfying $36a^3 \equiv 19 \pmod{p}$. Then for every integer $n \equiv a \pmod{p}$,

$$p \mid 36n^3 - 19.$$

This is immediate: $36n^3 - 19 \equiv 36a^3 - 19 \equiv 0 \pmod{p}$ whenever $n \equiv a \pmod{p}$.

Theorem 1 (Multi-prime combination). *Let $p_1, \dots, p_k$ be distinct primes, each $\equiv 2 \pmod{3}$, and let $a_1, \dots, a_k$ be the corresponding roots from Construction 1. Let $M = p_1 p_2 \cdots p_k$ and let A be the unique solution modulo M of the simultaneous system $n \equiv a_i \pmod{p_i}$ ($1 \leq i \leq k$), given by the Chinese Remainder Theorem. Then for every $n \equiv A \pmod{M}$,*

$$M \mid 36n^3 - 19, \quad \text{i.e.} \quad \frac{36n^3 - 19}{M} \in \mathbb{Z}.$$

Proof. For each i , $n \equiv A \equiv a_i \pmod{p_i}$, so $p_i \mid 36n^3 - 19$ by Construction 1. The p_i are distinct primes, so their product M divides $36n^3 - 19$. $\square$

So Theorem 1 answers the original question in the strong (infinite family) sense: $n \equiv A \pmod{M}$ and $x = M$ is a divisor of $36n^3 - 19$ for *every* such n , and M, A can be made as large as desired by taking more/larger primes p_i .

3 Corrected elliptic-curve specialization

Let $M, A \in \mathbb{Z}$, with $M \neq 0$, satisfy

$$M \mid 36A^3 - 19, \quad Q = \frac{36A^3 - 19}{M}.$$

Set

$$x = M, \quad n = A + Mt, \quad t \in \mathbb{Z}.$$

Expanding by exact integer arithmetic gives

$$\frac{36(A + Mt)^3 - 19}{M} = Q + 108A^2t + 108AMt^2 + 36M^2t^3.$$

Consequently the remaining square condition is

$$\boxed{y^2 = 36M^2t^3 + 36M(3A + M)t^2 + 12(3A + M)^2t + (6A + M)^2 + Q.} \quad (1)$$

This is a genuine integral-point problem on a genus-one curve (provided the cubic is nonsingular), and divisibility has already been solved structurally.

Write the right side of (1) as

$$P(t) = at^3 + bt^2 + ct + d,$$

where

$$\begin{aligned} a &= 36M^2, & b &= 36M(3A + M), \\ c &= 12(3A + M)^2, & d &= (6A + M)^2 + Q. \end{aligned}$$

The substitution

$$X = at, \quad Y = ay$$

gives the integral Weierstrass model

$$Y^2 = X^3 + bX^2 + acX + a^2d. \quad (2)$$

An integral point on (2) maps back to an admissible integer pair (t, y) only if $a \mid X$ and $a \mid Y$. Every recovered point must then be checked in the original equation. The associated cubes are

$$u = y - M, \quad v = 2M + 6(A + Mt), \quad w = -M - y,$$

and exact verification requires $u^3 + v^3 + w^3 = 114$.

3.1 Rank-guided modulus selection

The scalable CRT theorem creates infinitely many candidate pairs (M, A) . They should not be ordered merely by the number of digits in M . For each singleton prime or selected CRT product, we instead:

1. construct (2) and its minimal model;
2. compute the global root number;
3. use root number -1 as a heuristic priority for odd rank;
4. compute algebraic-rank bounds, analytic rank, and Mordell–Weil generators for the most promising curves;
5. determine integral points and retain only the sublattice $a \mid X, Y$;
6. verify the original square equation and the sum of cubes exactly.

Root number and rank are filters, not certificates of an integral solution. Positive rank need not produce a point in the required sublattice, while rank-zero curves can still have torsion points that must be checked. The script `rank_guided_elliptic_search.py` implements the construction, PARI rank bounds and analytic rank, bounded t -searches, and exact recovery tests.

4 Sanity check on small primes

Before trusting the construction at 30-digit scale, Lemma 1 and Construction 1 were checked by brute force for every prime $p < 200$ with $p \equiv 2 \pmod{3}$: for each such p , the formula's output a was compared against an exhaustive search over $\{0, \dots, p-1\}$ for roots of $36t^3 \equiv 19 \pmod{p}$, confirming in every case that (i) a root exists, (ii) it is unique, and (iii) it matches the closed-form value from the lemma. Representative small case: $p = 17$ gives the unique root $a = 1$, and indeed $36 \cdot 1^3 - 19 = 17$.

5 Two 30-digit primes ($k = 2$)

Two primes $p_1, p_2 \equiv 2 \pmod{3}$, each exactly 30 digits, were generated and fed through Construction 1:

$$p_1 = 432256083118531044858259323599$$

$$a_1 = 205375962562755822717169031496$$

$$p_2 = 508951041752918591462038694231$$

$$a_2 = 11998528096759146724786261916$$

Direct check: $36a_1^3 - 19 \equiv 0 \pmod{p_1}$ and $36a_2^3 - 19 \equiv 0 \pmod{p_2}$, both verified exactly.

Combining via CRT (Theorem 1, $k = 2$) gives modulus $M = p_1p_2$ (60 digits) and representative A (59 digits):

$$M = 219997183807212542923931096695598007560918353360263243457369$$

$$A = 61583691402182783047260797510703276439292047987969866884962$$

so that for every $n \equiv A \pmod{M}$,

$$\frac{36n^3 - 19}{M} \in \mathbb{Z}.$$

For the representative $n = A$ itself, $36A^3 - 19$ is a 178-digit integer whose exact quotient by M is the 119-digit integer:

$$Q = 38219282687467843755039989619336413202799418213239056237824742757700405657626958942597846112998216708072139112513223381$$

This was verified by exact big-integer arithmetic: $36A^3 - 19 = M \cdot Q$ precisely (no rounding, no floating point).

6 Three 30-digit primes ($k = 3$)

Pushing Theorem 1 to $k = 3$ with three fresh 30-digit primes $p_1, p_2, p_3 \equiv 2 \pmod{3}$:

$$p_1 = 779282871203804318581334453057$$

$$a_1 = 540863547358444129637446662643$$

$$p_2 = 917485502986111972709840010173$$

$$a_2 = 765515263147582646243402496698$$

$$p_3 = 422843868764454355911216773723$$

$$a_3 = 173885648024960625137792429386$$

each individually verified to satisfy $36a_i^3 \equiv 19 \pmod{p_i}$. The combined modulus is the 90-digit integer $M = p_1p_2p_3$:

$M = 302325220948348183420434660024015348902086281522777512680498475446030651529604772341579503$

with CRT representative (90 digits):

$A = 182470001039324181181024873011044406335174068803296028244172097475931272694261226503431598$

By Theorem 1, for *every* integer $n \equiv A \pmod{M}$,

$$x = M \quad \text{divides} \quad 36n^3 - 19.$$

Concretely, at $n = A$, $36A^3 - 19$ is a 270-digit integer, and $(36A^3 - 19)/M = Q$ is the following 180-digit integer:

$Q = 723440045918937868278733134032344285097446574084321029094017717823428606222435878944285733479052368954298584972235671037551447033227302246880204085670050855668916609015702186192131$

verified exactly via $36A^3 - 19 = M \cdot Q$ in arbitrary-precision arithmetic.

7 Two 40-digit primes ($k = 2$)

The same construction repeated with 40-digit primes $p_1, p_2 \equiv 2 \pmod{3}$:

$p_1 = 7729484335457653901640057298531371241781$

$a_1 = 7668575607239450973459863267707132263860$

$p_2 = 2486598372481845396683104279916570951657$

$a_2 = 609530524018264138310326718615033307496$

each verified individually against $36a_i^3 \equiv 19 \pmod{p_i}$. Combining via CRT gives the 80-digit modulus $M = p_1 p_2$ and 80-digit representative A :

$M = 19220123168672920512532048525080739795893456098633704192461052168447373009581117$

$A = 12055573306960757229878249326862599227290509643013675419894826308035075337947215$

so that for every $n \equiv A \pmod{M}$, M divides $36n^3 - 19$. At $n = A$, $36A^3 - 19$ is a 239-digit integer, and $(36A^3 - 19)/M = Q$ is the following 160-digit integer:

$Q = 3281783589609439060274407871106333299264698047304666391611865457187494168830153506599538715008344851754116620345469289412464296994115391491979199807003772632893$

verified exactly via $36A^3 - 19 = M \cdot Q$.

8 Three 40-digit primes ($k = 3$) — the largest instance found

Finally, three 40-digit primes $p_1, p_2, p_3 \equiv 2 \pmod{3}$:

$p_1 = 6719198563363963000306859976745602011477$

$a_1 = 3490701603819271511450476386979369043828$

$p_2 = 5014753468477740779475587342292599756573$

$a_2 = 2189794512725854244347716309834217468129$

$p_3 = 4350736286376564530047381650448847768399$

$a_3 = 525176878515189528938215354529508172272$

each individually verified to satisfy $36a_i^3 \equiv 19 \pmod{p_i}$. The combined modulus is the 120-digit integer $M = p_1 p_2 p_3$:

$M = 146598599970416865193503095382122376155797697327993857757772703444309911094348084964151091360030057229085267028040668079$

with CRT representative (119 digits):

$A = 78645806109075982274155955035418384113009237358084530484254201316396650239497999707497613326168293812811504563595780749$

By Theorem 1, for *every* integer $n \equiv A \pmod{M}$,

$$x = M \quad \text{divides} \quad 36n^3 - 19.$$

At $n = A$, $36A^3 - 19$ is a 359-digit integer, and $(36A^3 - 19)/M = Q$ is the following 240-digit integer:

$Q = 119453638517902053794963958283357629777707983376380601474787235288607383318513130562995313813671450899537612665803296173072136255334374568105736108037208680482571587733372866341630091424872047846333343491478053473403358730772112480717305455$

verified exactly via $36A^3 - 19 = M \cdot Q$ in arbitrary-precision arithmetic.

9 Discussion and honest audit

- **Proved, general, and scalable.** Lemma 1 and Theorem 1 are elementary and fully rigorous for *any* choice of distinct primes $p_i \equiv 2 \pmod{3}$ (about half of all primes, by Dirichlet). Nothing here is specific to 30-digit or 40-digit numbers — the same construction works verbatim for 100-digit or 1000-digit primes; these sizes were simply the scales requested. Using k primes of d digits each yields a modulus M (and hence divisor x) of size $\approx dk$ digits.
- **What “largest” means here.** In this construction, $x = M$ is chosen to be a product of the p_i so that divisibility holds across an entire residue class of n . Increasing the number or size of the primes makes M larger, but the elliptic-curve selection criteria in Section 3 are more relevant than a notion of “largest.” The $k = 3$, 40-digit instance in §7 (x a 120-digit number, n a 119-digit number) is the largest computed here; $k = 4, 5, \dots$ or larger primes would push this arbitrarily further at the cost of more/larger primes, with no change in the method.
- **If instead a single fixed integer n is wanted** (not a whole congruence class) with $x \mid 36n^3 - 19$ for some unrelated, independently-chosen large x , that reduces to integer factorization of $36n^3 - 19$ for a specific n , which is a fundamentally different (and much harder, non-constructive) problem — one has to search and factor rather than construct. The CRT/cube-root method above sidesteps that difficulty entirely by *choosing* the primes first and reverse-engineering n , which is why it scales cleanly to 30-digit (or larger) moduli.
- **Corrected next step.** The CRT output is now fed into the fixed specialization $x = M$, $n = A + Mt$, which yields the elliptic family (1). Independent affine congruence classes for n and a variable x are not claimed to imply $x \mid 36n^3 - 19$.
- **Verification.** Every numerical claim in this document was checked with exact Python and SymPy arbitrary-precision arithmetic: primality of each p_i , the congruence $p_i \equiv 2 \pmod{3}$, the root identity $36a_i^3 \equiv 19 \pmod{p_i}$, the CRT combination, and the final exact divisions $36A^3 - 19 = M \cdot Q$, for the $k = 2$ and $k = 3$ instances at both the 30-digit and 40-digit scales (four instances total). The closed-form cube-root formula was additionally cross-checked against brute force for all primes $p < 200$ with $p \equiv 2 \pmod{3}$.